Models that remember too much



AI security · Level 3 · Lesson 2

Today we will

- Explain memorisation and membership inference.
- Remove identifiers from text before it is used.
- Weigh privacy against usefulness, and ask the right questions.

Remember

- Rare details can be memorised and repeated word for word.
- Once data trains a model, it is hard to take out.
- Keep it out; redact what must go in; test with a canary.

Black it out

1. Warm-up (5 min): What could a register tell a stranger?
2. Teach (8 min): How models memorise; what counts as an identifier.
3. Black it out (12 min): Groups remove every identifier.
4. Swap and check (8 min): Find what the other group missed.
5. Wrap-up (7 min): Agree a rule, then the exit ticket.

Use only made-up names and numbers, never a real register. Hide two or three rare details (“the only pupil who walks from the river village”) so groups learn that identifiers are not only names and numbers.

Let’s discuss

1. Which identifiers were hardest to spot, and why?
2. Why can't we just delete the file after a model has learned from it?
3. When is it worth making a model less useful to protect privacy?

Exit ticket

- Write three kinds of identifier, and one question to ask before any pupils' data goes to an AI service.